

Afsana Bhuiyan

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Work Experience

Riverside Health & Rehabilitation Center

Remote

Software Engineer (Freelance)

06/2026 – Present

- Built and maintain [CareTrack](#), a React, Express, and PostgreSQL app in production serving up to 100 residents. It tracks admissions and automates physician documentation tasks and deadlines.
- Designed security around HIPAA's technical safeguards, including role-based access control, audit logging of every patient-data access, JWT auth in httpOnly cookies with CSRF protection, account lockout, and idle-session timeouts.
- Containerized the API with Docker, deployed to Azure Container Apps and Static Web Apps, and built a GitHub Actions pipeline that lints, tests, builds, and security-audits every push.

Margin Research

New York, NY

Software Engineer (Contract)

02/2026 – 06/2026

- Shipped Reagent, a SaaS platform that flags malicious open-source contributors, from research prototype to v1 for government customers in four months as one of four engineers.
- Designed the FastAPI/Pydantic REST endpoints and implemented an asynchronous Celery/Redis pipeline that dispatched code-analysis jobs to 6 internal tools (each its own microservice), indexing results in Elasticsearch and storing raw artifacts in S3.
- Created a reusable TypeScript visualization tool for graph-based analysis, adaptable to multiple data formats and per-node analytics which was adopted across two internal projects.
- Containerized application services with Docker, built a CI/CD pipeline in GitHub Actions, and wrote pytest unit tests with 85% test coverage.

Pieces

New York, NY

Software Engineer Intern

09/2025 – 01/2026

- Built a full-stack Flask and JavaScript internal tool using the Gemini API to generate synthetic patient records for training and testing the company's AI document-summarization product.
- Applied prompt engineering and built a data pipeline to auto-repair malformed LLM responses, enabling reliable generation of 200+ clinical notes per patient.
- Modeled patient record data models with physician consultants to ensure clinical realism, and added batch processing with JSON exports for downstream model training.

ISO New England

Holyoke, MA

Software Engineer Intern

06/2023 – 12/2023

- Built an automated reporting service in Java (Spring Boot, Apache Camel, Thymeleaf) that queries Oracle (PL/SQL) for the latest power-pricing data and emails each team member a daily chart dashboard.
- Tested an internal Java tool that calculates total transfer capability (how much power can reliably move between grid regions), measuring run times with SQL queries and analyzing results in Excel.
- Presented findings to the engineering team and at the company-wide intern fair.

Projects

[El Principito](#) - Language App | C#, ASP.NET Core, EF Core, SQLite, React Native

08/2026 – Present

- Designed a layered REST API with dependency-injected services for stories and vocabulary.
- Developing a React Native and TypeScript (Expo) app with a tap-to-translate reading view.

Education

Rensselaer Polytechnic Institute | Troy, NY

05/2025

B.S. in Computer Science, Data and AI Specialization

Skills

Languages: Python, TypeScript, JavaScript, SQL, C#, Java, C/C++, HTML/CSS

Frameworks & Libraries: FastAPI, Flask, Node.js, Express.js, ASP.NET Core (.NET), Entity Framework Core, Celery, React, React Native, pandas, NumPy, SpringBoot

Databases & Cloud: PostgreSQL, Redis/Valkey, Elasticsearch, MongoDB, SQLite, AWS (S3), Azure, Supabase

DevOps & Tools: Docker, GitHub Actions, Git, Linux, Swagger/OpenAPI, pytest